

Section: **4.3 – FATIGUE ANALYSIS**
 Chapter: **4310**
 Title: **STRESS CONCENTRATION FACTOR**
 Revision: **0.1**
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1 INTRODUCTION

This section deals with the stress concentration factor (SCF or K_t) calculation.

The stress concentration factor is a measure of the stress increase that appears at the root of geometrical notches, such as holes, grooves, shoulders, etc. The theoretical stress concentration factor, $K_{t_{net}}$, is defined as the ratio between the peak stress at the root of the notch and the nominal stress which would be present if a stress concentration did not occur (Ref. 2):

$$K_t = K_{t_{net}} = \frac{\sigma_{max}}{\sigma_{net}}$$

Other reference stress can be used depending on the particular aspects of the analysis as the stress in the gross section (gross stress) or the bearing stress for loaded holes. The stress concentration factor shall be defined always coherently such as the product of the reference stress and the stress concentration factor is always the notch stress:

$$\sigma_{max} = K_{t_{net}} \cdot \sigma_{net} = K_{t_{gross}} \cdot \sigma_{gross} = K_{t_{bearing}} \cdot \sigma_{bg}$$

Where σ_{net} , σ_{gross} and σ_{bg} are the net stress, gross stress and bearing stress respectively and σ_{max} is the peak stress, also known as σ_{notch} .

It is important to note that the stress concentration factor, K_t , is a theoretical factor based on assumptions in theory of elasticity. No plasticity is assumed.

The K_t values for a wide variety of geometries and conditions can be obtained from different engineering manuals. The most commonly used are "Stress Concentration Factors" by R.E. Peterson. (Ref. 1) and ESDU items (Ref. 10).

2 INFLUENCE OF K_t ON THE CRACK INITIATION LIFE

According to the previous explanation, the SCF is a measurement of the increase of the stress in a local area due to stress concentrating features such as holes, grooves, notches, etc. Since the stress is higher in these areas than in the rest of the structure, these locations are prone to suffer fatigue crack initiation failures.

Figure 1 shows the S-N curves for different stress concentration factors. It can be seen how the SCF influences on the crack initiation life for a given stress level.

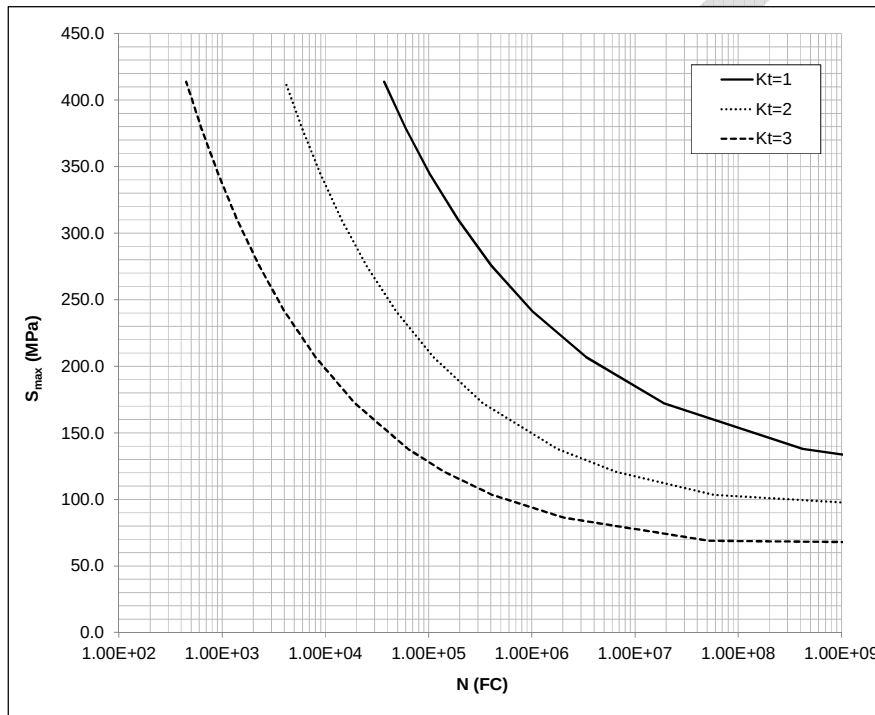


Figure 1. S-N curves for different K_t for 2024T3 ($R = 0$)

The S-N curves are given for particular values of $K_{t,net}$. Therefore, for the crack initiation analyses, the stress concentration factor should be always expressed in terms of net stress.

Some remarks about how the stress concentration factor influences on the crack initiation life_[HH1]:

- For the same reference stress, the higher the K_t , the lower the crack initiation life:

$$\sigma_{ref} = const. \quad K_t \uparrow \Rightarrow \text{Crack Initiation Life} \downarrow$$

- Since the stress concentration factor is calculated without considering plasticity, the notch stress could be higher than the ultimate material stress, without failure of the component. This is because the load redistribution due to the material plasticity. The real stress in the structure would be lower than the notch stress:

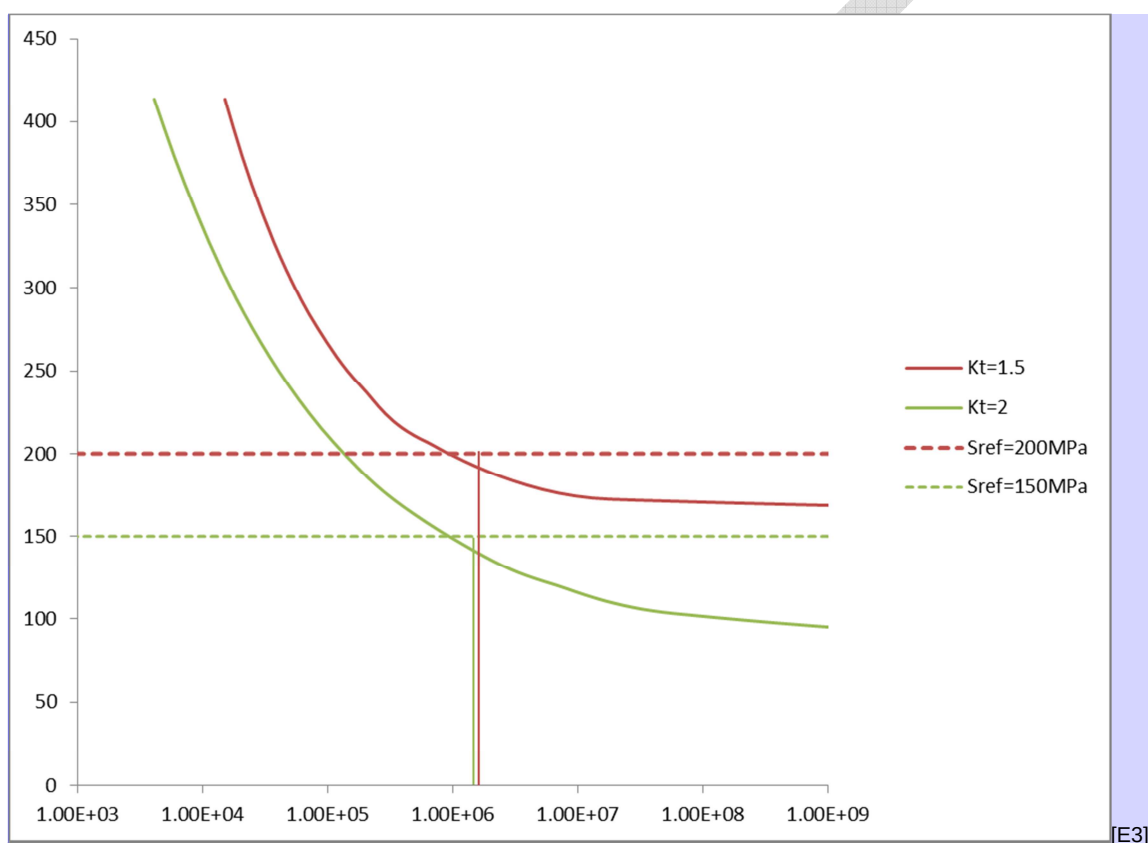
$$\sigma_{max,real} \leq \sigma_{max} = K_t \cdot \sigma_{ref}$$

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- For those cases where the maximum stress is slightly higher than the yield stress, the higher the K_t , the higher crack initiation life for the same notch stress:

$$\sigma_{max} = \text{const.} \quad K_t \uparrow \Rightarrow \text{Crack Initiation Life} \uparrow$$

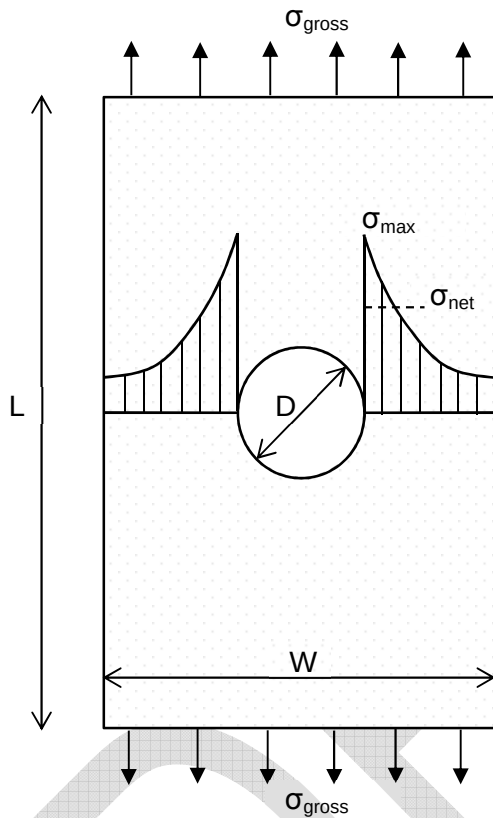
- For those cases where the maximum stress is lower than the yield stress, the K_t has little influence on the crack initiation life_[HH2].



3 Kt CALCULATION FOR HOLES

This chapter deals with the stress concentration factor calculation for holes, under different geometric and load configurations.

The presence of a hole in an axial tension loading plate causes a stress distribution, see Figure 2, with a stress concentration at the hole edge. The stress concentration factor for holes, $K_{t_{net}}$, is defined as the ratio between the peak stress (σ_{max}) at the hole edge and the nominal stress in the net cross-sectional area (σ_{net}). It could be also useful to calculate the stress concentration factor as the ratio between the peak stress at the hole edge and the far-field/gross stress (σ_{gross}):



$$\sigma_{net} = \frac{\sigma_{gross}}{1 - \frac{D}{W}}$$

$$K_{t_{net}} = \frac{\sigma_{max}}{\sigma_{net}} \quad K_{t_{gross}} = \frac{\sigma_{max}}{\sigma_{gross}}$$

Figure 2. Stress concentration around a hole

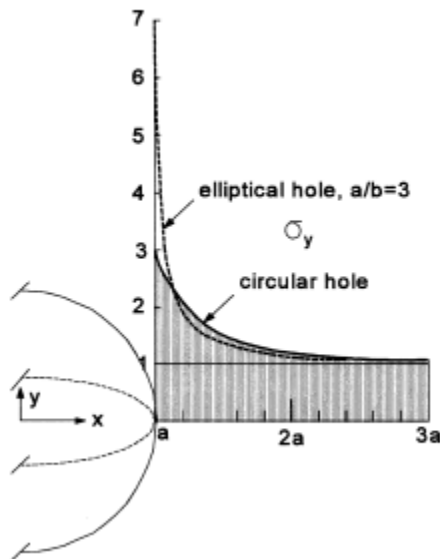
The stress concentration factors, $K_{t_{net}}$ and $K_{t_{gross}}$, are related to each other in next equation:

$$K_{t_{net}} = K_{t_{gross}} \cdot \left(1 - \frac{D}{W}\right) = K_{t_{gross}} \cdot \frac{1}{SAF_{net}}$$

Where the Stress Adjustment Factor (SAF_{net}) is the ratio between the net cross-sectional area and the gross section area.

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Sometimes it could be important to know the stress distribution along the cross section, that is, how fast the stress decreases away from the root of the notch. For a circular hole, the stress distribution along the x-axis is given by:



$$\sigma(x) = \sigma_{gross} \cdot \left[1 + \frac{1}{2} \cdot \left(\frac{a}{x} \right)^2 + \frac{3}{2} \cdot \left(\frac{a}{x} \right)^4 \right]$$

Figure 3. Stress gradients at the root of the notch (from Ref. 2)

The general solution for a circular hole in an infinite plate under uniaxial tension loading was firstly solved by Kirsch in 1898 (Ref. 3). Kirsch's solution states that the hoop stress around the edge is:

$$\sigma_{\varphi} = \sigma_{gross} \cdot (1 - 2\cos(2\varphi))$$

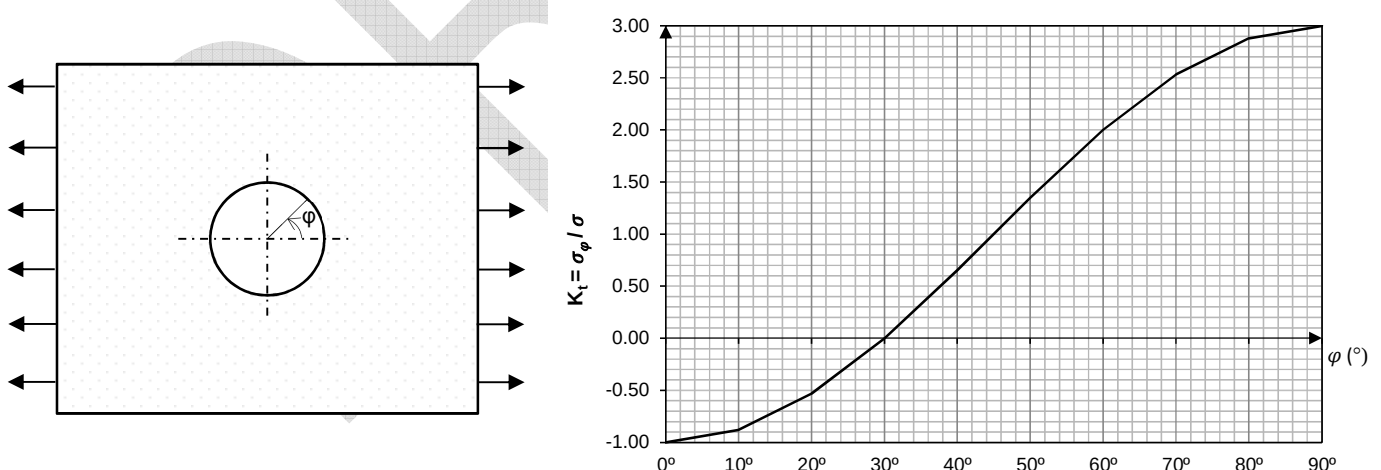


Figure 4. Stress concentration factor for a circular open hole in an infinite width panel [HH4][E5]

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The stress around the hole edge varies from 3σ at $\varphi = 90^\circ$ (perpendicular to load direction) to $-\sigma$ at $\varphi = 0^\circ$ (load direction). Therefore, the stress concentration factor for an open hole in an infinite width panel is:

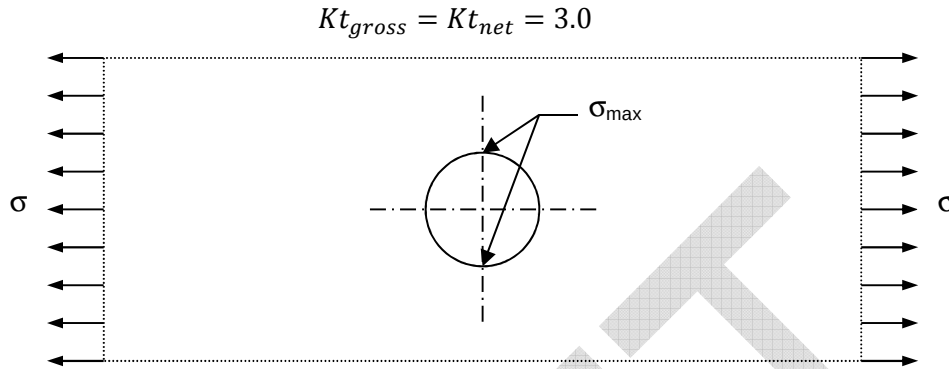


Figure 5. Open hole in an infinite width panel. Uniaxial tension

The stress concentration factor for a circular hole in an infinite width plate in biaxial tension loading, as the shown in Figure 6, can be obtained by superposition of σ_1 and σ_2 uniaxial stress distributions.

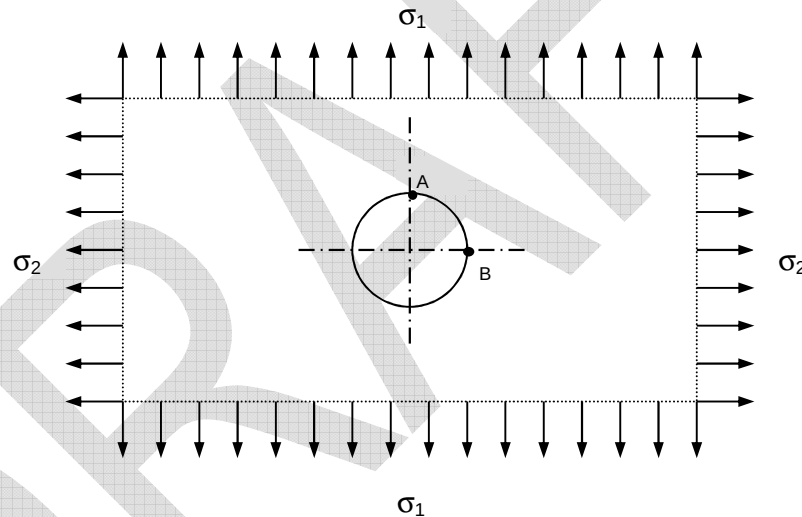


Figure 6. Centred open hole in a finite width panel - Biaxial load

The SCF at the points A and B are:

$$Kt_A = 3 \cdot \frac{\sigma_2}{\sigma_1} - 1 \quad Kt_B = 3 - \frac{\sigma_2}{\sigma_1}$$

It is interesting to note that:

- If σ_1 and σ_2 are both either tension or compression the stress concentration factor is less than 3, which is the stress concentration factor caused by uniaxial stress.
- For equal biaxial stress, $\sigma_1 = \sigma_2$, the stresses at A and B are $\sigma_A = \sigma_B = 2\sigma_1$.
- If σ_1 and σ_2 are the same magnitude but opposite sign ($\sigma_1 = -\sigma_2$) the stress concentration factor is 4 (pure shear).

The exposed above are the basic stress concentration factors for open holes (only by-pass load) in infinite width plates, but they can be affected by geometrical effects and load effects. Therefore, the analysis of the SCF for holes needs the known of the stress state and the geometry of the hole.

The geometrical effects on the SCF can be summarizing in two groups:

- Proximity to an edge or to other holes
- 3D geometrical effects:
 - o Thickness
 - o Countersunk
 - o Pin bending
 - o Shadow [HH6][E7]
 - o Filling

Regarding the stress state, it must be distinguished between:

- By-pass tension load
- By-pass shear load
- Bending load
- Axial load transfer
- Shear load transfer

Once all the contributions have been considered, they can be combined as follows:

- Combination of geometrical effects:

$$K_t = K_{t_1} \cdot K_{t_2} \cdot \dots K_{t_n}$$

- Combination of load cases:

$$K_t = K_{t_1} + K_{t_2} + \dots + K_{t_n} \text{ [HH8][E9]}$$

Chapter 3.4 exposes with detail how to combine all the effects for the stress concentration factor calculation of a joint.

Note that next chapters show the solution used in Airbus DS for 2D plates with circular holes. Other solutions such as elliptical holes, plates with pattern of holes, etc., can be found in Ref. 1.

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Figure 1 shows the influence on the stress concentration factor of the different configurations of a hole without load transfer. It can be seen that the K_t increases when effects as the proximity to an edge are added, while it decreases if the hole is filled up or the rivet is installed with interference:

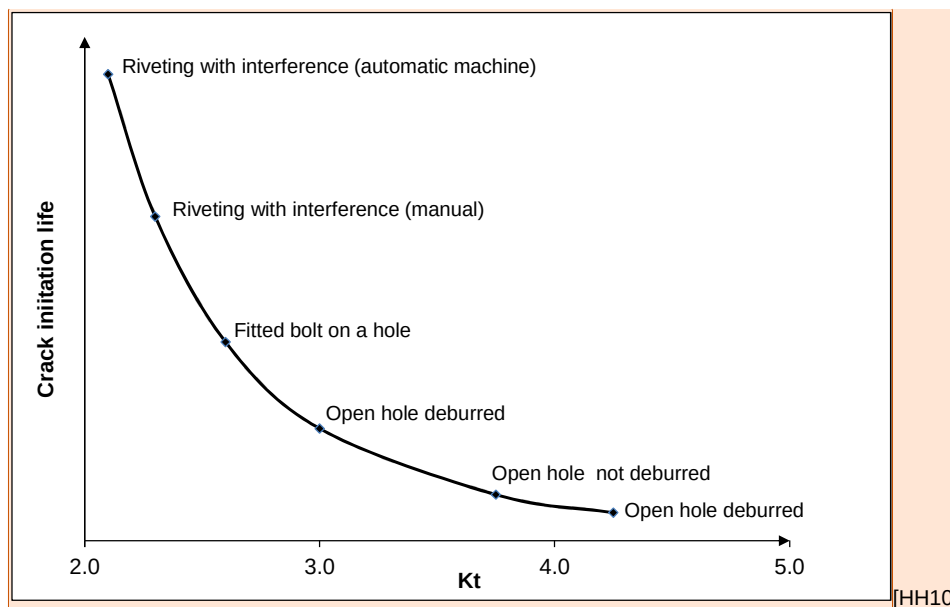


Figure 7. Influence on the SCF of different configurations of a hole (no load transfer)

Figure 8 shows the stepwise to be followed to calculate the stress concentration factor for a typical joint.

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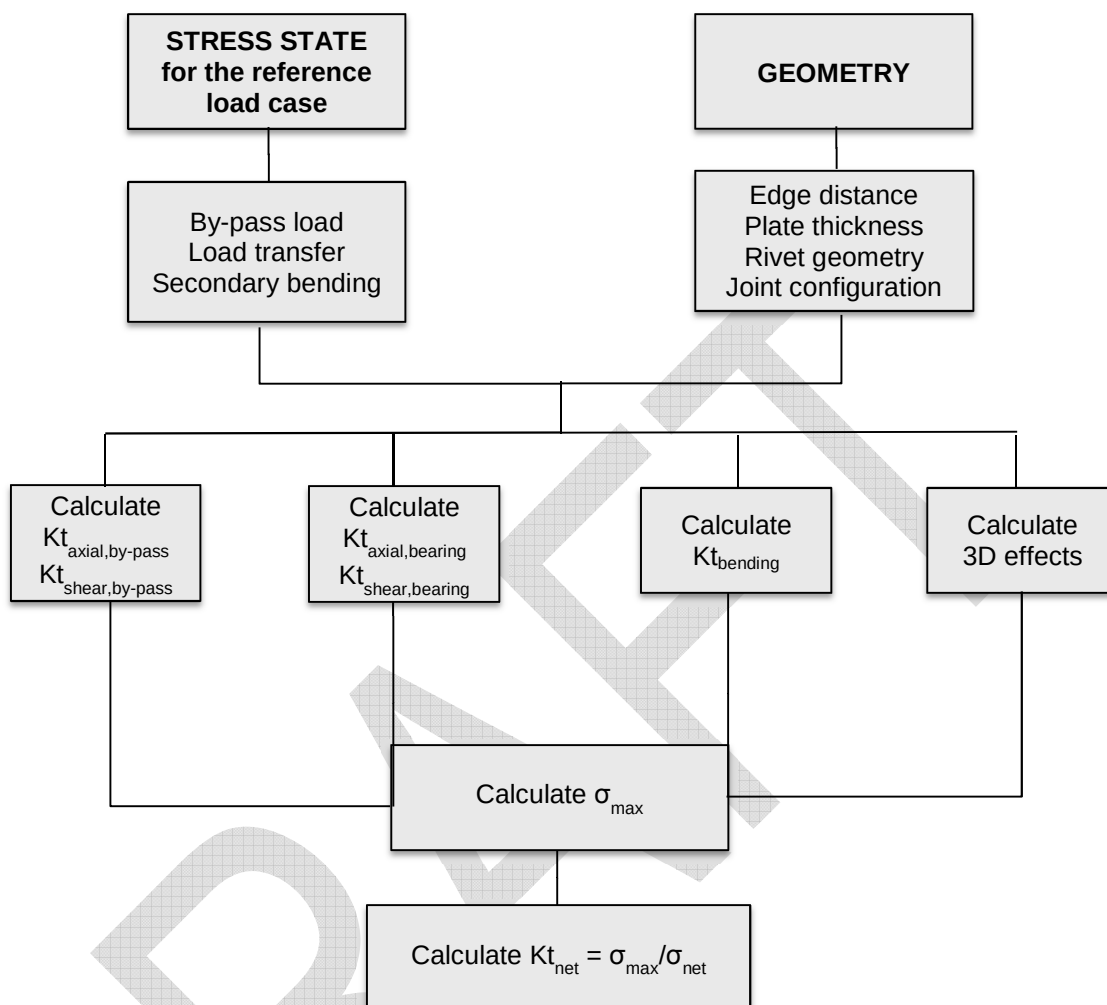


Figure 8. Stress concentration factor calculation

3.1 Geometrical effects

3.1.1 Proximity to an edge

Circular hole under by-pass tension load

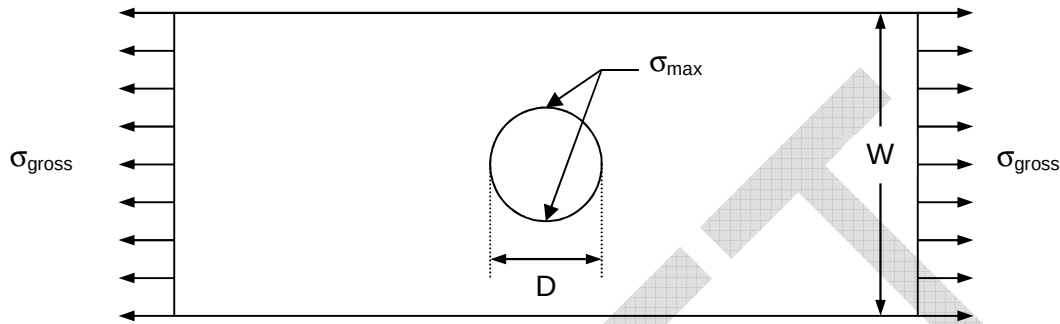


Figure 9. Centred open hole in a finite width panel - Axial tension load

The stress concentration factor of a centred hole in a finite width panel can be obtained from the following empirical equation (Heywood, 1952):

$$Kt_{net} = 2 + \left(1 - \frac{D}{W}\right)^3$$

$$Kt_{gross} = \frac{2 + \left(1 - \frac{D}{W}\right)^3}{1 - \frac{D}{W}} \Rightarrow SAF_{net} = \frac{1}{1 - \frac{D}{W}}$$

The maximum stress around the hole is obtained as:

$$\sigma_{max} = \sigma_{gross} \cdot Kt_{net} \cdot SAF_{net}$$

Although the length of the plate is not in the stress concentration factor equation, the Kt value depends also on this dimension, but this influence is negligible if the length is more than twice the plate width (Ref. 2).

Figure 10 shows the influence of edge proximity on the Kt value; Kt_{gross} varies from 3.0 when $W \gg D$ ($D/W = 0$) to 4.25 when the edge distance is one diameter ($D/W = 0.5$).

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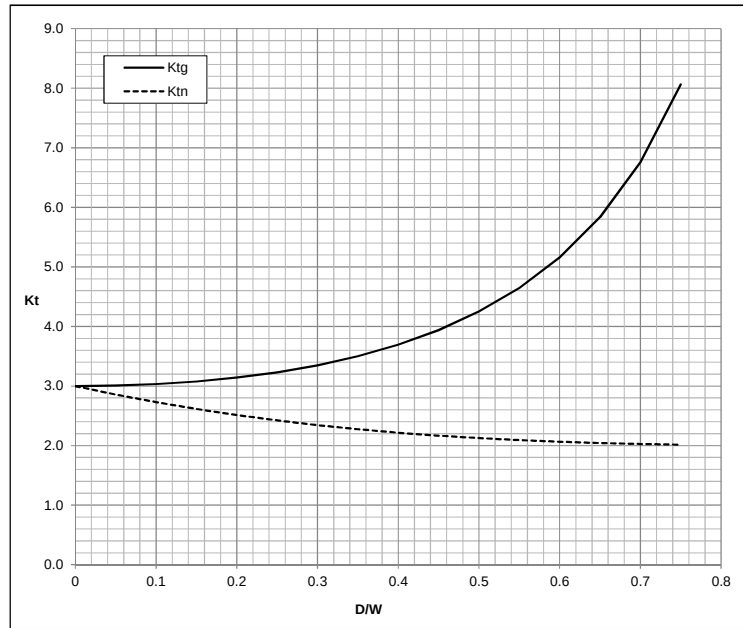


Figure 10. Stress concentration factor for a centred open hole in a finite width panel – Axial load

For the non-centred holes the stress concentration factor can be obtained graphically from Chart 4.3 from Ref. 1 and follows the next equation:

$$Kt_{net} = 3 - 3.140 \cdot \frac{D}{2c} + 3.667 \cdot \left(\frac{D}{2c}\right)^2 - 1.527 \cdot \left(\frac{D}{2c}\right)^3$$

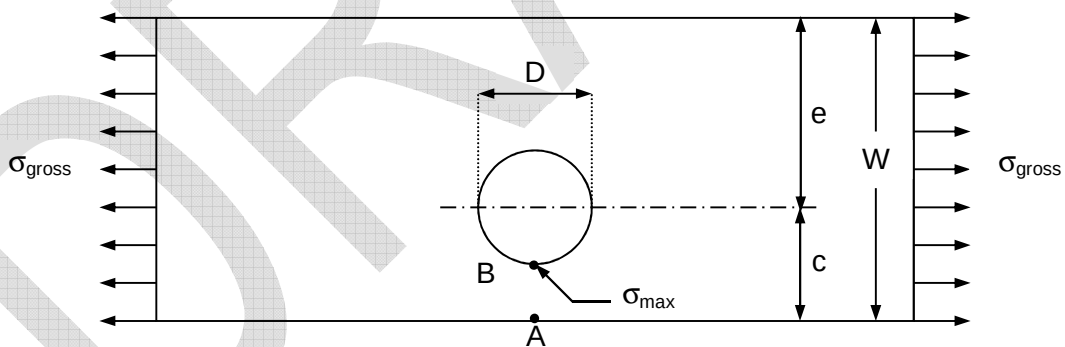


Figure 11. Eccentrically located open hole in a finite width panel - Axial load

It can be seen in Ref. 1 (see Figure 12) that all the Kt_{net} curves obtained for this case are very close for typical design values ($c \geq 2D$), so that it is a reasonable approximation for practical purposes to take the curve for the centred located hole ($e/c = 1$, $W = 2c$) as the unique solution:

$$Kt_{net} = 2 + \left(1 - \frac{D}{2 \cdot c}\right)^3$$

STRESS CONCENTRATION FACTOR

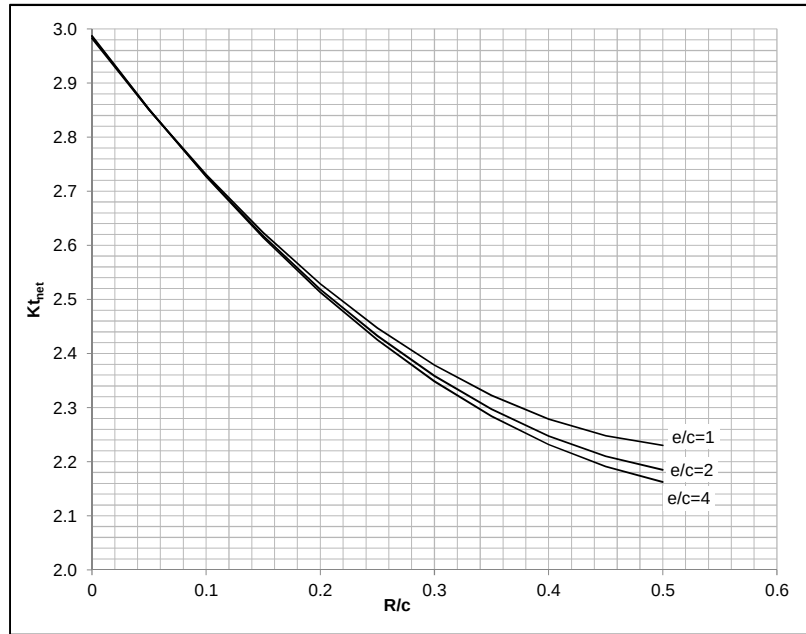


Figure 12. Stress Concentration Factor for non-centred holes (from Ref. 1)

Similarly, the gross factor, Kt_{gross} , shall be calculated as the factor for a centred located hole ($e/c = 1$):

$$Kt_{gross} = \frac{2 + \left(1 - \frac{D}{2 \cdot c}\right)^3}{1 - \frac{D}{2 \cdot c}}$$

The reference stress for this case is the average stress on the weakest cross section (A-B in Figure 11). Then, the maximum stress around the hole and the reference stress are defined as:

$$\sigma_{max} = Kt_{net} \cdot \sigma_{net,A-B} = Kt_{net} \cdot \sigma_{gross} \cdot SAF_{net} = Kt_{gross} \cdot \sigma \cdot SAF_{gross}$$

$$\sigma_{net} = \sigma \cdot SAF_{net}$$

According to Ref. 1, the SAF_{net} for an eccentric hole is:

$$SAF_{net} = \frac{\sqrt{1 - \left(\frac{D}{2 \cdot c}\right)^2}}{\left(1 - \frac{D}{2 \cdot c}\right) \cdot \left(1 - \frac{c}{W - c} \cdot \left(1 - \sqrt{1 - \left(\frac{D}{2 \cdot c}\right)^2}\right)\right)}$$

And the SAF_{gross} can be derived as:

$$SAF_{gross} = \frac{SAF_{net}}{1 - \frac{D}{2 \cdot c}}$$

Note that Kt_{gross} calculated in this way is different from the one obtained from Chart 4.3 of Ref. 1:

$$Kt_{gross}(\text{Chart 4.3} - \text{Ref. 1}) = Kt_{gross} \cdot SAF_{gross}$$

Circular hole under by-pass shear load

The maximum stress location, ϕ , for an open hole in a finite width panel under shear stress depends on the ratio W/D , being perpendicular to the maximum principal stress ($\phi=135^\circ$ - 45° in Figure 13) for infinite width panels:

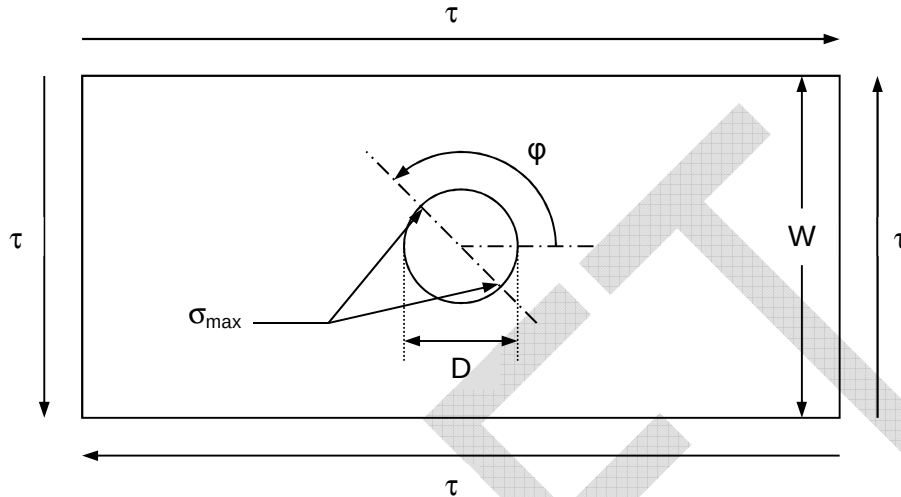
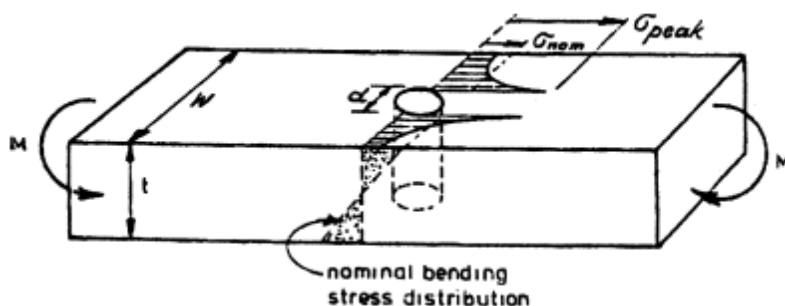


Figure 13. Open hole in a finite width panel under shear loads

There is not an analytical solution for the stress concentration factor for this case. For this reason, Airbus DS has developed a database for practical purposes with the stress concentration factor solution (K_t and ϕ) for shear loaded panels with the parameter W/D as variable parameter. This database is implemented in the Airbus DS legacy tool AGUJERO (see Ref. 4).

Circular hole under bending load

For plates with holes loading in bending, the stress concentration factor K_t is the ratio between the maximum stress at the hole edge and the bending stress:



$$K_{t,net} = \frac{\sigma_{peak}}{\sigma_{net}} = \sigma_{peak} \cdot \frac{1}{\frac{6M}{(W-D) \cdot t^2}}$$

$$K_{t,gross} = \frac{\sigma_{peak}}{\sigma_{gross}} = \sigma_{peak} \cdot \frac{1}{\frac{6M}{t^2}}$$

Figure 14. Stress distribution in a beam with a transverse hole loading in bending (from Ref. 2)

The stress concentration factor for bending of a plate with a circular hole can be obtained graphically from Chart 4.91 and Chart 4.92 from Ref. 1 for infinite and finite width plates respectively.

3.1.2 3D geometrical effects

Thickness effect

In a plate with a transverse hole under an arbitrary stress state, the stress distribution varies across the plate thickness. The stress is lower at the surface and higher at the mid-thickness. The stress distribution across the thickness depends on the thickness to diameter ratio (t/D), where D is the diameter of the hole and t the plate thickness, as well as on the distance from the mid-thickness.

Historically, the Airbus legacy methods have obtained the thickness effect factor (F_{th}) to be applied on the SCF from Figure 15.

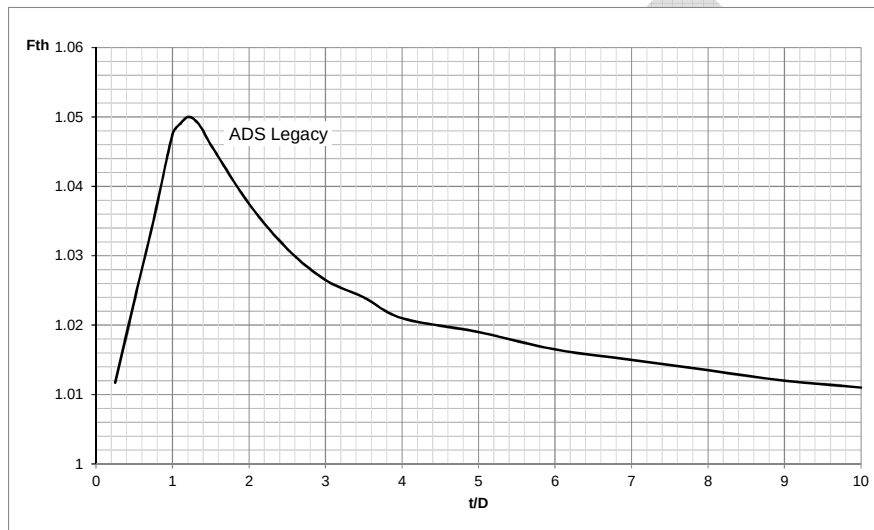


Figure 15. Effect of t/D on SCF (ADS Legacy Method)

Recently, other authors as Shivakumar et al. (Ref. 6) have derived expressions obtained from FE models results to calculate the thickness effect:

$$F_{th} = 1 + \frac{0.5 \cdot \left(2 \cdot \frac{t}{D}\right)^{0.6}}{10 + \left(2 \cdot \frac{t}{D}\right)^{1.6}}$$

Both solutions are similar and show the same trend but give different results for the factor F_{th} . Since the expression derived by Shivakumar in Ref. 6 has been correlated recently with FE models and it is slightly more conservative, it will be assumed as the new approach for the calculation of the thickness factor for future ADS projects. Figure 16 shows a comparison between both approaches.

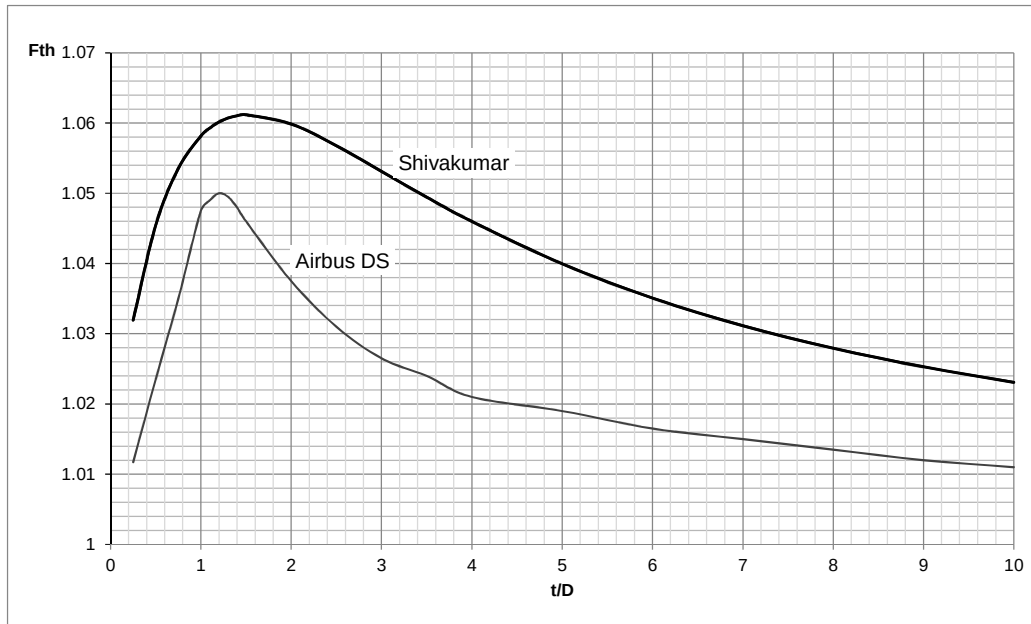


Figure 16. Effect of t/D on SCF: Shivakumar (Ref. 6) vs Airbus DS approach

Countersunk effect

Figure 17 shows the geometry of a countersunk hole and the location where the maximum stress concentration appears. The plate thickness in this case is divided into two portions: the sinking thickness (c) and the straight shank thickness (b), so that $t = c + b$. The straight shank diameter is D and the countersunk angle is θ .

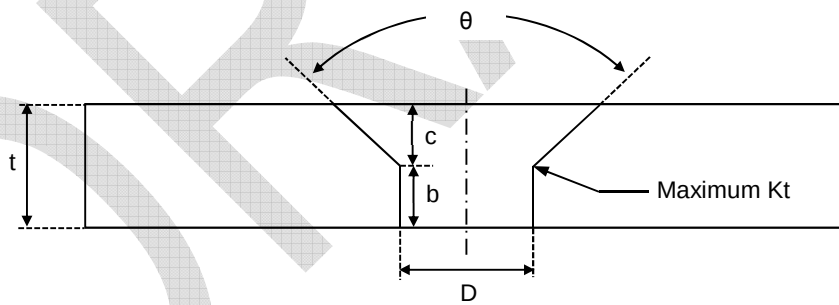


Figure 17. Countersunk hole

Several studies to obtain the influence of the countersunk on the SCF have been performed by:

- Young and Lee (1993). It is the one proposed by Peterson's 3rd edition (Ref. 1):

$$F_{CSK} = 1 + 0.72 \cdot \frac{c}{t}$$

- Northrop Corporation. This is the approach historically used by Airbus DS. It is an analytical formula obtained from 3D FE model simulation, valid in the interval $c/t < 0.7$:

$$F_{CSK} = 1 + 0.4786 \cdot \frac{c}{t}$$

- Shivakumar and Newman (Ref. 5)
- Shivakumar et al. (2006). It was obtained from 3D FEM models. It is also the option recommended by Peterson's 3rd edition (Ref. 1):

$$F_{CSK} = F_{th} \cdot F'_{CSK} = \left[\frac{0.5 \cdot \left(2 \cdot \frac{t}{D}\right)^{0.6}}{10 + \left(2 \cdot \frac{t}{D}\right)^{1.6}} \right] \cdot \left[1 + 0.4 \cdot \left(\frac{c}{t}\right) \cdot \left(2 \cdot \frac{c}{t}\right)^{0.6} - 0.1 \cdot \left(\frac{c}{t}\right)^2 \cdot \left(2 \cdot \frac{t}{D}\right)^{0.3} \right]$$

- ESDU approach. It is based on the Shivakumar 3D FE models and it is available through a table interpolator.

Other members of the aeronautical industry such as McDonnell Douglas Corporation, Boeing Company or Airbus have developed their own formulas to take into account this effect.

Airbus DS performed a benchmark in Ref. 7 to compare different approaches and their field of validity. The recommendation is the use of Shivakumar's formula in the region with $D/t < 1.23$ and maintain Northrop solution in the region with $D/t > 1.23$ for the calculation of the countersunk effect:

$$\frac{D}{t} \leq 1.23 \Rightarrow \text{Northrop: } F_{CSK} = 1 + 0.4786 \cdot \frac{c}{t}$$

$$\frac{D}{t} > 1.23 \Rightarrow \text{Shivakumar: } F_{CSK} = \left[\frac{0.5 \cdot \left(2 \cdot \frac{t}{D}\right)^{0.6}}{10 + \left(2 \cdot \frac{t}{D}\right)^{1.6}} \right] \cdot \left[1 + 0.4 \cdot \left(\frac{c}{t}\right) \cdot \left(2 \cdot \frac{c}{t}\right)^{0.6} - 0.1 \cdot \left(\frac{c}{t}\right)^2 \cdot \left(2 \cdot \frac{t}{D}\right)^{0.3} \right]$$

Pin bending effect

In a plate with a loaded hole, the stress distribution varies across the plate thickness due to the fastener flexibility (pin bending). This effect appears only if there is a load transfer through the rivet.

Figure 18 shows the maximum stress concentration area. In a single shear joint the K_t is maximum in the contact surface and minimum in the free surface. Likewise, in a double shear joint the K_t is maximum in the contact surfaces and minimum in the middle plane of the joint.

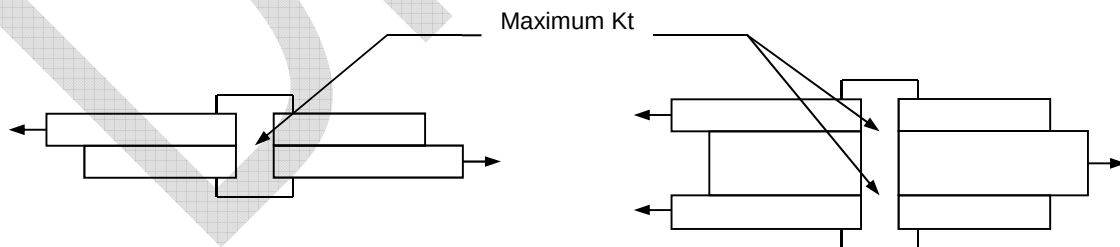


Figure 18. Pin bending effect on the stress concentration factor

The pin bending effect depends mainly on the plate thickness and on the ratio between the stiffness of the plate and the rivet (E_r/E_p).

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Historically, the Airbus DS legacy methods have obtained the pin bending effect factor (F_{pb}) to be applied on the SCF from Figure 19 for different ratios E_r/E_p , where E_r and E_p are the fastener and the panel young modulus respectively.

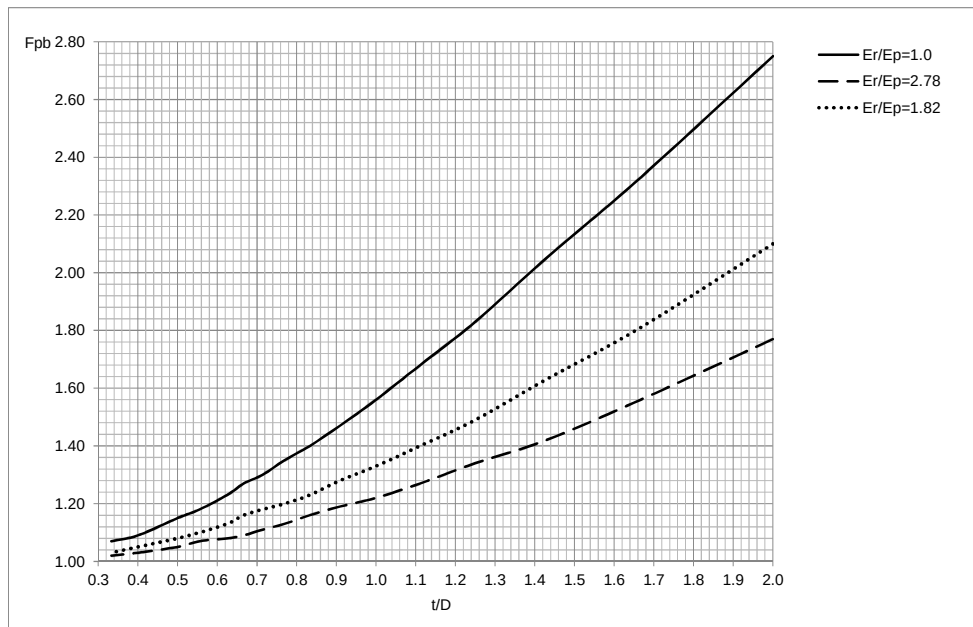


Figure 19. Effect of pin bending on SCF (ADS Legacy Method)

Other sources of literature such as NIU (Ref. 8) and ESDU (Ref. 10) propose other values for the pin bending effect which are quite different from the values historically adopted by Airbus DS when $t/D > 0.5$. A comparison between different approaches for the calculation of the pin bending factor can be found in Ref. 9 and it is shown in Figure 20.

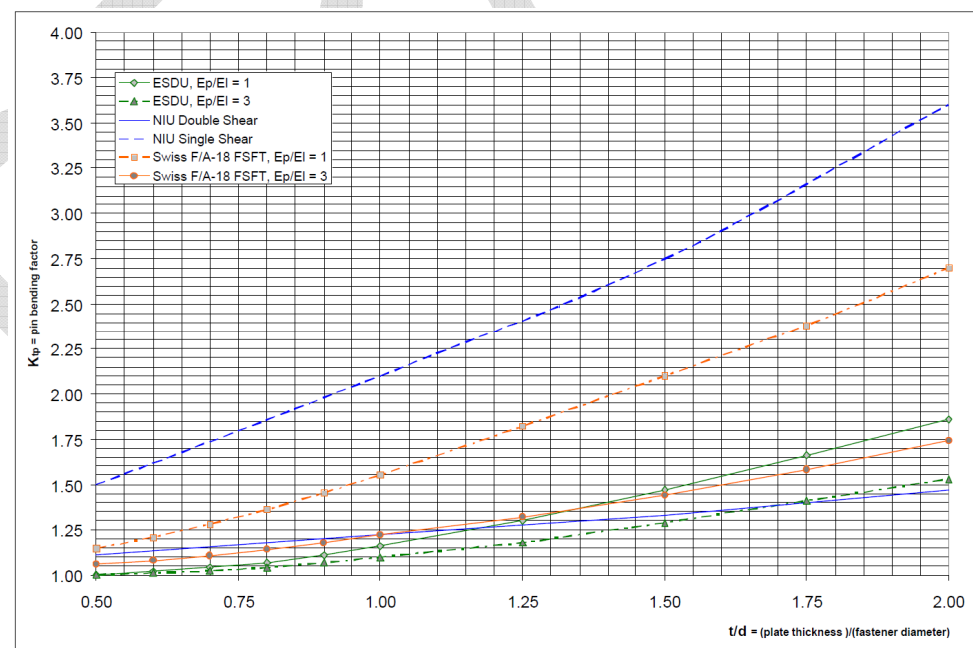


Figure 20. Comparison between different approaches for the pin bending factor calculation (from Ref. 9)

In Figure 20 the curves "Swiss F/A – 18 FSFT" correspond with the method used in Airbus DS.

It is important to note that ESDU (Ref. 10) and Airbus DS Legacy method considers the influence of different young modulus of the fastener and the plate while NIU (Ref. 8) does not consider it but it takes into account the joint configuration (single shear or double shear).

Ref. 9 concludes in its analysis that detailed 3D FE models are required to assess the stress distribution across the joint thickness. On the other hand, both ESDU and NIU states that the pin bending effect results provided should be considered just as a guide because the data was obtained for a few set of geometry configurations.

Since it is not possible to use 3D FE models for the analysis of each joint, and the approach of ESDU and NIU don't give enough confidence to be applied on an arbitrary geometry, the Airbus DS method can be remain as the standard method to calculate the pin bending effect until a further analysis could be performed. **UNDER DISCUSSION.**

Shadow effect

When comparing the stress concentration of one hole row with the one obtained for an infinite holes row loaded axially parallel to the row, there are differences in the maximum stress concentration. This is the so-called shadow effect. The shadow effect appears as a stress concentration reduction factor when holes are added to the row and its effect is higher as the distance between holes (pitch) is reduced.

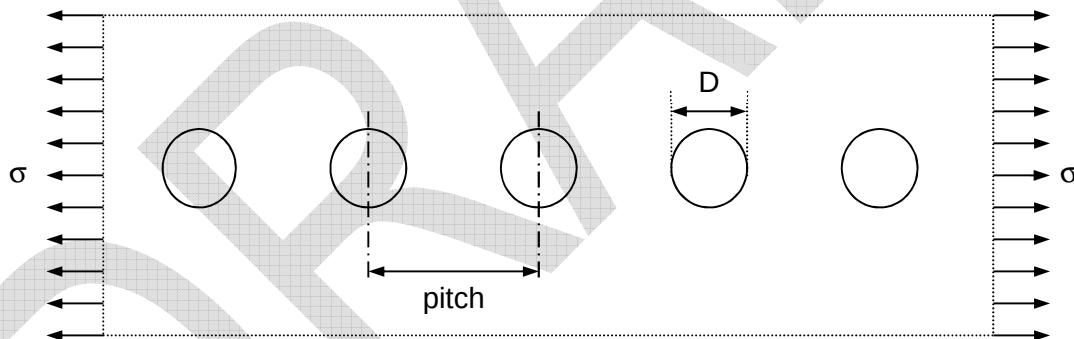


Figure 21. Infinite row of circular holes – Uniaxial tension load parallel to the axis of the holes

The shadow effect can be obtained from Charts 4.22 and Chart 4.33 of Ref. 1 for the extreme and intermediate holes respectively:

In practice, the shadow effect is typically ignored in order to be conservative in the SCF calculation. However, if a more detailed fatigue assessment needs to be performed, the interaction between holes can be considered.

Filling effect

The holes are usually plugged with some type of rivet, bolt or other fastener installation, depending on the component to be analysed, which have a beneficial effect on the crack initiation life because of the reduction of the stress concentration factor due to the hole deformation prevention.

The stress concentration factor for a hole with a rivet with no load transfer under axial or shear by-pass can be obtained from the FE models developed for the deployment of the Airbus DS tool AGUJERO (see Ref. 4). It was obtained for different ratios E_r/E_p where E_r is the rivet young modulus and E_p is the panel young modulus.

Note that the filling factor was obtained for non-loaded holes but not for holes with load transfer, for which the SCF reduction due to the hole filling is expected to be lower. Therefore, not considering the filling effect in the initial design is a usual practice in Airbus DS in order to be conservative ($E_r/E_p = 0$) but its consideration can be a recovery solution if early findings appear in test or in service.

3.2 Kt calculation for loaded holes

The estimation of the stress concentration factor for pin-loaded holes can be obtained from the FE models that were developed for the deployment of the Airbus DS tool AGUJERO (see Ref. 4).

3.2.1 Axial load transfer

For an axially pin-loaded centred hole in a finite width panel the maximum stress location is located perpendicular to the load.

The stress concentration factor, calculated as the ratio between the maximum stress around the hole and the bearing stress, for centred holes is implemented in tabular form in Airbus DS tool AGUJERO. The complete process that was followed to obtain them is presented in Ref. 4.

$$Kt_{bg} = \frac{\sigma_{max}}{\sigma_{bg}} = \frac{D \cdot t}{P} \cdot \sigma_{max}$$

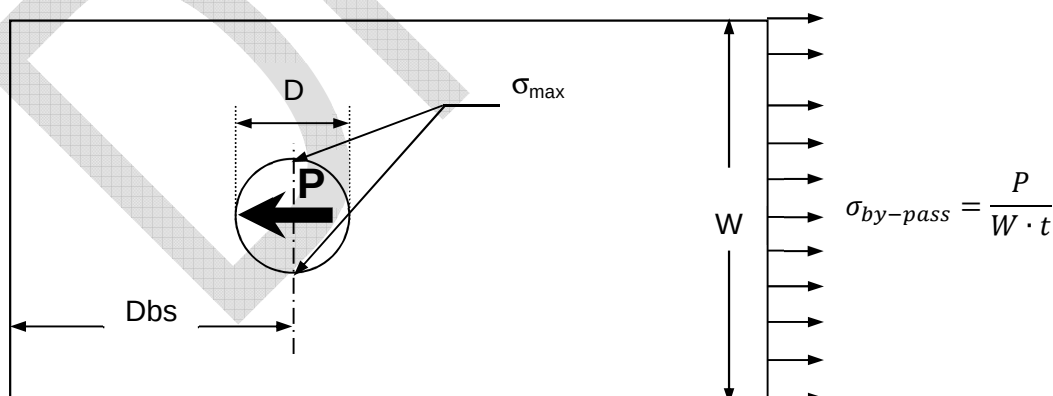
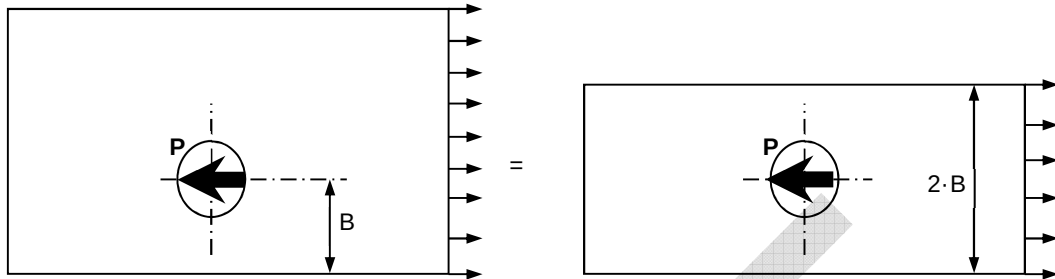
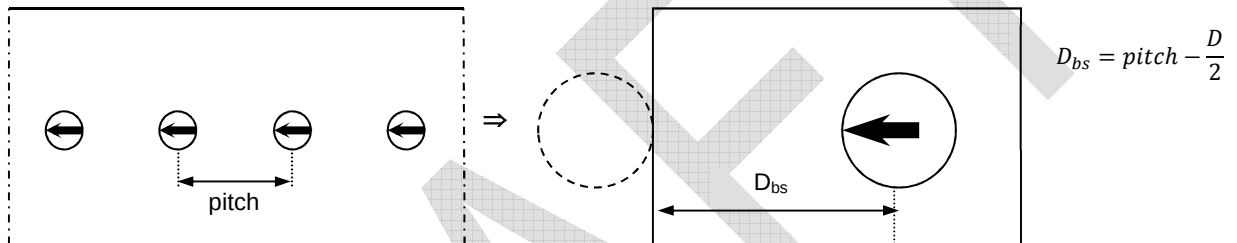


Figure 22. Centred hole in a finite-width panel. Axial bearing hole

For non-centred holes the SCF can be obtained using the same assumptions as in the analysis of the centred hole, but considering the total width as twice the distance to the nearest edge (2B):



In the case of a row of pin-loaded holes, the stress concentration factor can be calculated for each hole independently using the same assumptions as in the analysis of the centred isolated hole, but considering the distance D_{bs} as $D_{bs} = \text{pitch} - D/2$



3.2.2 Shear load transfer

Figure 23 shows the maximum stress location and the stress state for a shear joint in a finite width panel. The maximum stress is found in the hole points perpendicular to the maximum principal stress direction and closer to the edge where the load is reacted.

The stress concentration factor, calculated as the ratio between the maximum stress around the hole and the bearing stress, is implemented in tabular form in Airbus DS tool AGUJERO. The complete process that was followed to obtain them is presented in Ref. 4.

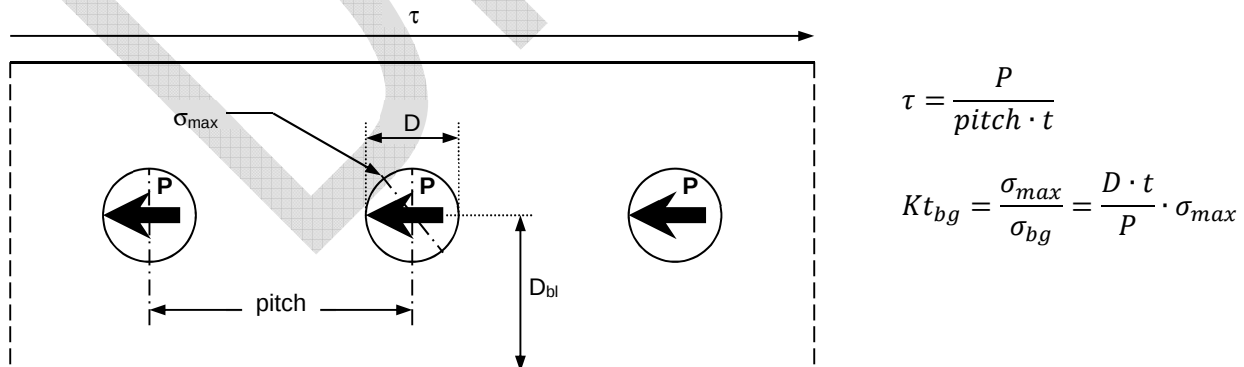


Figure 23. Pin-loaded hole in a finite width panel. Shear joint [HH11]

3.3 Secondary bending

The secondary bending appears in joints loaded in tension as a consequence of the eccentricity in the joint; the neutral line of the tension load action changes its relative position and it induces a secondary out-of-plane bending loading. Secondary bending stresses can also occur in plate elements with non-symmetric thickness steps

This secondary bending is the cause of a change of the local stress that must be considered through an additional K_t or a variation factor of the far stress.

The secondary bending can be obtained either from DFEM, analytical procedures or also from strain gauges measurements.

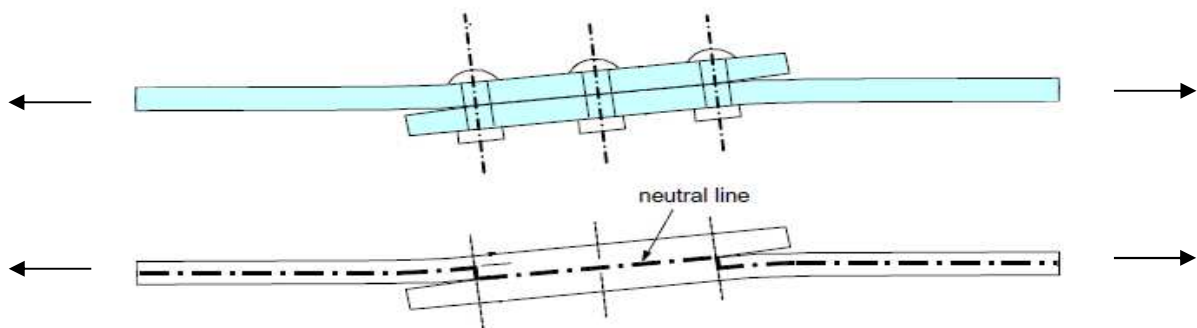


Figure 24. Secondary bending in a riveted tension joint

The secondary bending depends directly on the total eccentricity (the sum of the plates' thicknesses) but also on the geometry of the joint. For example, lower secondary bending stress can be obtained by increasing the overlap of the joint, staggering of the thickness or increasing the row spacing.

The secondary bending depends directly on the total eccentricity but also on the geometry of the joint. Figure 25 shows four different joint geometries and their bending factor (k_b) which is the ratio of the bending stress and the applied tensile stress. Ref. 13 concludes that, when possible, symmetric double strap joints in which eccentricities do not occur are recommended (Figure 25, a). However, in this case the crack detectability should be taken into account. The second best option is the strap joint with staggered straps (Figure 25, b). The lap joint (Figure 25, c) and the single strap joint (Figure 25, d) exhibit the higher secondary bending effects [HH12].

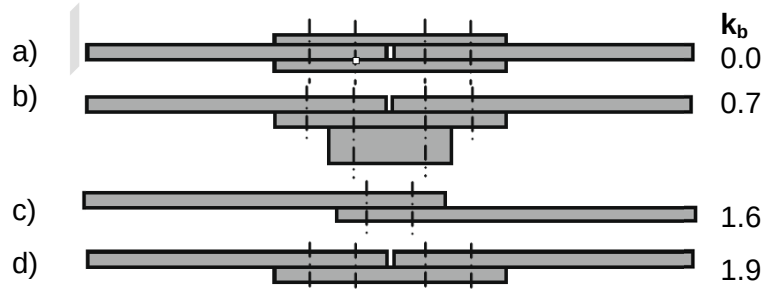


Figure 25. Secondary bending factor for different riveted joint geometries (from Ref. 13)

3.4 Kt calculation for joints

As it was said in section 1, the total stress concentration factor can be calculated as the combination of the individual effects of geometry and load cases.

The stress concentration factor for a general joint, loaded with axial and shear by-pass stress and axial and shear load transfer and secondary bending at the hole, the stress concentration factor is calculated taking into account all the contributions:

$$Kt_{net} = \frac{\sigma_{notch}}{\sigma_{net}} = \frac{1}{\sigma_{net}} \cdot [F_{th} \cdot F_{csk} \cdot (\sigma_{notch,axial\ bypass} + \sigma_{notch,shear\ bypass} + F_{pb} \cdot \sigma_{notch,axial\ transfer} + F_{pb} \cdot \sigma_{notch,shear\ transfer} + Kt_{bending} \cdot \sigma_{bending})]$$

Where:

- $\sigma_{notch,axial\ bypass}$ is the maximum stress due to the axial by-pass stress. It can be calculated as:

$$\sigma_{notch,axial\ bypass} = Kt_{axial\ bypass,net} \cdot SAF_{axial\ bypass,net} \cdot \sigma_{axial\ bypass,gross}$$

Where $Kt_{axial\ bypass,net}$ and $SAF_{axial\ bypass,net}$ can be calculated following the methodology exposed in section 3.1.1.

- $\sigma_{notch,shear\ bypass}$ is the maximum stress due to the shear by-pass stress (see section 3.2.2).
- $\sigma_{notch,axial\ transfer}$ is the maximum stress due to the axial load transfer. It can be calculated as:

$$\sigma_{notch,axial\ transfer} = Kt_{axial\ bearing} \cdot \frac{P_{axial}}{D \cdot t}$$

Where $Kt_{axial\ bearing}$ can be calculated following the methodology exposed in section 3.2.1.

- $\sigma_{notch,shear\ transfer}$ is the maximum stress due to the shear load transfer (see section 3.2.2).
- $Kt_{bending}$ is the bending stress concentration factor. It can be obtained using Chart 4.83 from Ref. 1.
- $\sigma_{bending}$ is the secondary bending stress. It can be obtained from detailed FEM or convenient hypothesis (see chapter 3.3)
- F_{pb} , F_{th} , F_{csk} are the pin bending, thickness and countersunk effect factors (see chapter 3.1.2.)

STRESS CONCENTRATION FACTOR

Depending on the surface to be analysed (free surface, countersunk surface, contact surface or middle surface) the values of the factors F_{pb} , F_{th} , F_{csk} and $\sigma_{bending}$ have to be chosen coherently.

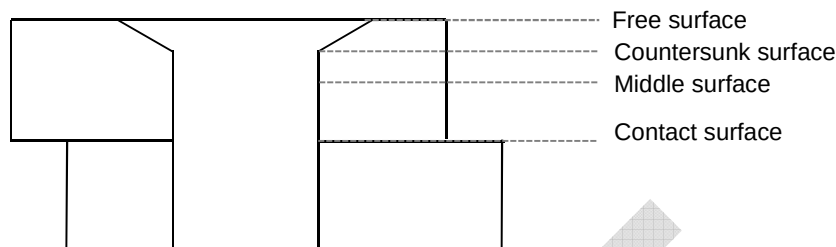


Figure 26. Joint surface to calculate the stress concentration factor

Surface	F_{th}	F_{csk}	F_{pb}	$\sigma_{bending}$
Middle	Yes	No	No	0.0
Countersunk	Yes/No (*)	Yes	No	$\sigma_{bending} (X=C)$
Contact	No	No	Yes	$+\sigma_{bending}$
Free	No	No	No	$-\sigma_{bending}$

Table 1. SCF parameters to consider depending on the surface

(*): It is important to note that the thickness effect in the countersunk surface is already considered when using Shivakumar approach ($t/D > 1.23$) for the countersunk effect calculation. If Northrop formula is used for the calculation of the countersunk effect, the thickness effect has to be considered additionally (see section 3.1.2).

Note that:

- If the shear load transfer and the axial load transfer are not in the same direction, the stress concentration factor cannot be calculated as the combination of their individual effects. Since the FE models used to calculate the SCF include non-linear contacts, the superposition principle is not applicable for this case.^[HH13]
- The maximum stress produced by the axial and shear bypass stress and axial and shear load transfer are not in the same point of the hole edge. A further analysis of the stress distribution around the hole is required in order not to be too conservative.

4 K_t CALCULATION FOR LUGS

The stress concentration factor for lugs in Airbus DS has been historically calculated using three methods:

- ESDU 81006: Stress concentration factors for axially loaded lugs with clearance-fit pins.

ESDU approach provides the maximum elastic stress concentration factors at the perimeter of the hole in lugs loaded axially by single central clearance-fit pins. This methodology covers only axially loaded straight lugs (see Figure 27). Nor tapered lugs and neither those lugs where the load is not aligned with the lug axis are covered by this approach.

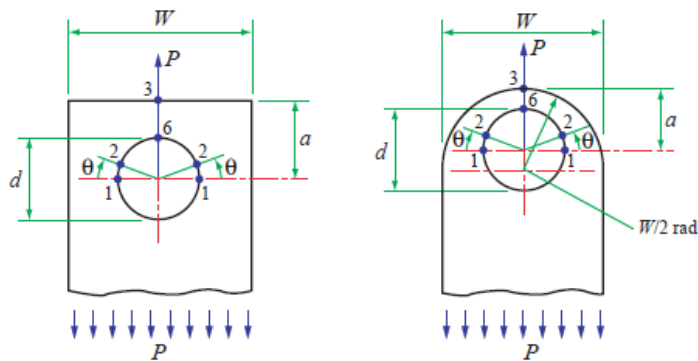


Figure 27. ESDU 81006 Squared and rounded-ended plain lugs geometry definition

- HSB 34112-02: Stress concentration factors for symmetrical isotropic lugs

HSB approach in Ref. 11 provides the stress concentration factor at the perimeter of the hole in lugs with no clearance where the material behaviour is elastic and the following geometry requirements are met:

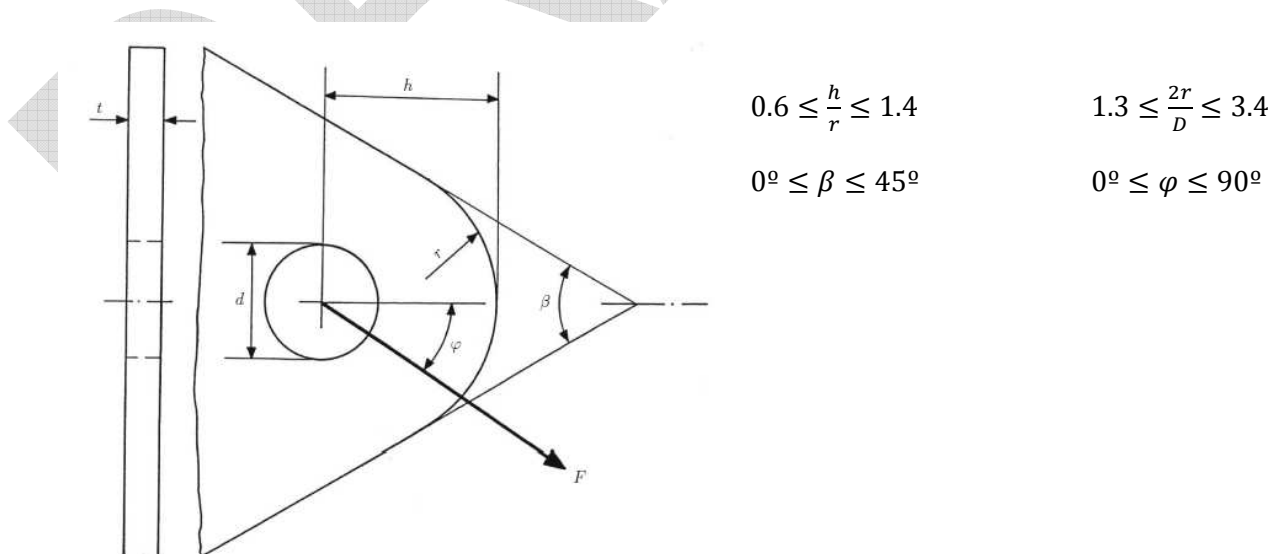


Figure 28. HSB 34112-02 Lug geometry definition

- Detailed Finite Element Models:

In some cases, a dedicated linear finite element model can be required to obtain the stress concentration factor of the lug accurately.

The stress concentration factor is calculated from the DFEM results as:

$$Kt_{bg} = \frac{\sigma_{max}}{\sigma_{bg}}$$

Where σ_{max} is the maximum stress around the hole and σ_{bg} is the bearing stress ($= P / D \cdot t$).

No correction to consider the pin bending is needed because the stress concentration factor obtained from LDFEM includes the effects due to the pin flexibility.

Due to the limitations of the ESDU methodology for the stress concentration factor calculation of lugs, it is recommended the use of the HSB approach when possible. A dedicated finite element model could be needed for those cases where a detailed analysis would be required or the geometry would be out of the range of applicability of HSB 34112-02. **UNDER DISCUSSION.**

4.1 HSB methodology

4.1.1 Stress concentration factor

The stress concentration factor in Ref. 11 is defined as the ratio between the maximum stress in the lug hole and the bearing stress:

$$Kt_{bg} = \frac{\sigma_{max}}{\sigma_{bg}} = \frac{\sigma_{max}}{\frac{F}{D \cdot t}}$$

The stress concentration factor, Kt_{br} , is calculated as:

$$Kt_{bg} = H \cdot Kt_{bg1}$$

Where H is the geometry factor and depends mainly on the ratio h/r and Kt_{bg1} is the stress concentration factor for the special case $h = r$

The expressions to calculate H and Kt_{bg1} can be found in Ref. 11.

4.1.2 Pin bending effect

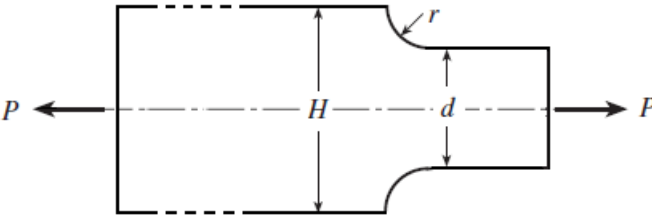
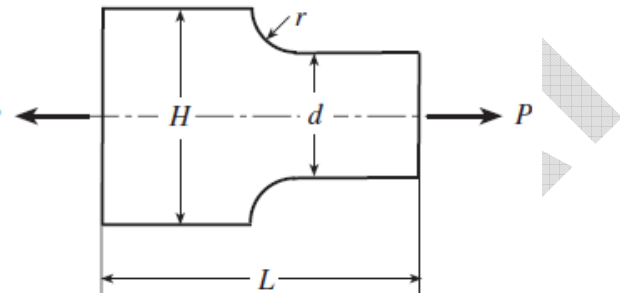
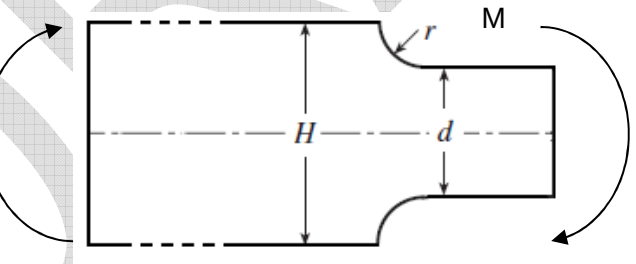
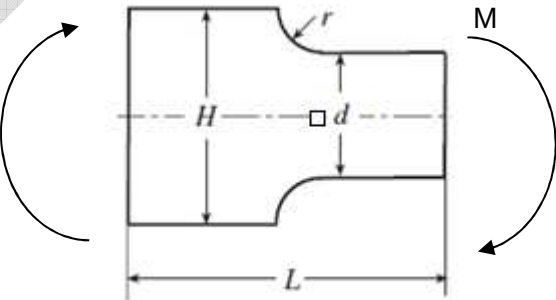
HSB approach does not consider the effect of the ratio t/D on the stress distribution along the lug thickness.

The pin bending has an important effect on the stress concentration factor for those cases with $t/D > 0.5$. Therefore, and although HSB does not states a specific methodology to consider the pin bending, this effect must be calculated and considered additionally to the stress concentration factor following the approach exposed in section 3.1.2.

5 K_t CALCULATION FOR NOTCHES

This chapter deals with the stress concentration factor calculation for notches, grooves and shoulders, under different geometric and load configurations.

Table 2 shows the reference to be used for each of the typical geometrical details. For other geometrical details, see Ref. 1, chapter 17 of Ref. 12, ESDU datasheets, HSB...

Detail	Reference
SCF for a stepped flat tension bar with shoulder fillets	
SCF for a stepped flat tension bar with shoulder fillets. Influence of element length (L)	
SCF for bending of a stepped flat bar with shoulder fillets	
SCF for bending of a stepped flat bar with shoulder fillets. Influence of element length (L)	

STRESS CONCENTRATION FACTOR

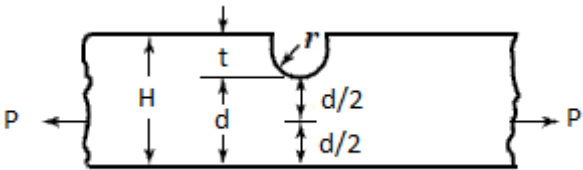
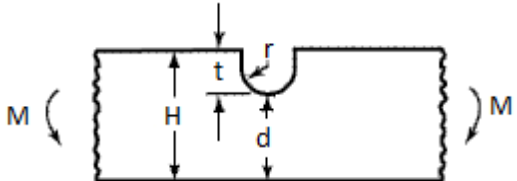
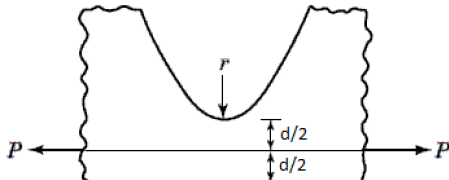
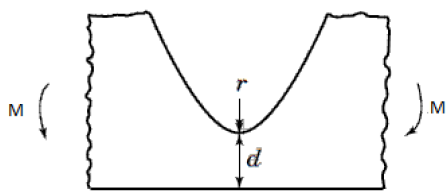
Flat tension bar with a U-shaped notch at one side.		Ref. 1, Chart 2.9
Bending of a thin beam element with a notch on one side		Ref. 1, Chart 2.30
SCF for tension loading of a semi-infinite thin element with a deep hyperbolic notch.		Ref. 1, Chart 2.8
SCF for bending of a semi-infinite thin element with a deep hyperbolic notch		Ref. 1, Chart 2.29

Table 2. SCF calculation for typical geometrical details

6 IMPROVEMENT IN THE FATIGUE BEHAVIOUR

The crack initiation life of a joint can be improved by reducing the stress concentration factor through technological procedures such as coldworking, the installation of the rivet with interference, applying shot-peening to the surface or eliminating roughness.

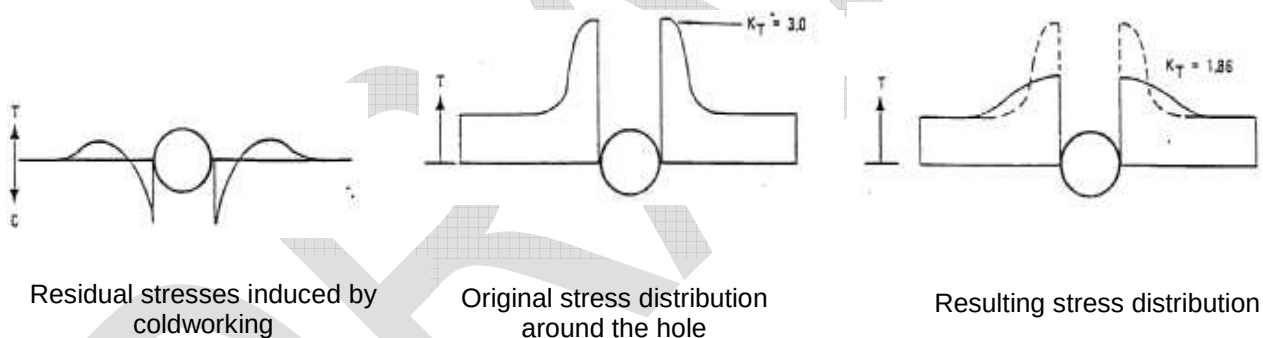
Shot-peening

It consists on the impacting of the surface with small steel balls. The plastic strains generated on the surface create residual compression stresses that improve the fatigue behaviour and limit the stress corrosion cracking. It is mainly used on aluminium or steel alloy elements and seldom on titanium alloys.

Although this technic is beneficial under the fatigue point of view, it is neglected when the crack initiation analysis is performed. However, it is used as an argument for a conservative analysis or as a recovery solution if early findings appear in test or in service.

Coldwork

Coldworking is a procedure that involves pulling an oversized, tapered mandrel through a hole. This produces a force that expands plastically the hole and creates a residual compressive stress zone around the hole. This zone extends approximately one radius from the edge of the hole and acts as a local crack initiation life enhancement by reducing the notch stress around the hole.



The improvement of the crack initiation life due to the coldwork is usually neglected in the initial analysis in Airbus DS. [HH14] However, it could be a recovery solution if early findings appear in test or in service. In the past, for some specific situations, an improvement factor on the crack initiation life between 2 and 3 was applied after detailed analyses considering the geometry and the specific location fatigue spectra. Therefore, if the consideration of the fatigue enhancement due to the coldwork would be required for a fatigue test correlation or for a repair analysis, contact your local stress authority.

Interference fit

The so-called interference fit refers to the installation of a rivet in a hole with a smaller diameter. This process produces mainly tensile tangential residual stresses around the hole. According to Ref. 14, when the rivet is installed with interference fit, mean notch stress in cyclic tensile loading increases but it considerably reduces the local stress amplitudes^[HH15], which is more influential than mean stress in the crack initiation.

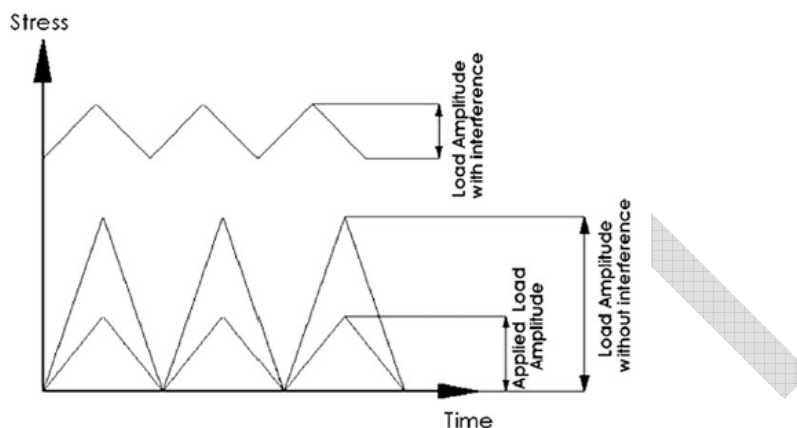


Figure 29. Stress amplitude variations before and after interference fit in a hole edge (from Ref. 14)

Although the rivet installation with interference fit improves the fatigue behaviour of the joint, its beneficial effect is neglected in Airbus DS in the initial crack initiation analyses. However, it is used as an argument for a conservative analysis or as a recovery solution recovery solution if early findings appear in test or in service.

PENDING TO DISCUSS:

- FASTENER FITTING CONDITIONS^[HH16]: CLEARANCE/INTERFERENCE FIT, PRELOAD. UNDER DISCUSSION.
- SURFACE TREATMENTS.

7 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
Airbus DS		Airbus Defence and Space
c		Countersunk depth
D		Diameter
D_{bs} , D_{bl}		Edge distance
E_p		Panel young modulus
E_r		Rivet young modulus
F_{csk}		Countersunk effect factor
F_{pb}		Pin bending effect factor
F_{sh}		Shadow effect factor
F_{th}		Thickness effect factor
k_b		Bending factor
Kt, SCF		Stress concentration factor
L		Length
P		Load
SAF		Stress Adjustment Factor
t		thickness
W		Width

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 R.E. Peterson	3 rd .	Stress concentration factors
Ref. 2 Schijve J.	2 nd	Fatigue of structures and materials
Ref. 3 Kirsch G.	1898	Die Theorie der Elastizität und Bedürfnisse der Festigkeitslehre
Ref. 4 NT-T-ADP-03016	A	Manual teórico del AGUJERO
Ref. 5 Shivakumar K., Newman J.C.	1992	Stress Concentrations for Straight-Shank and Countersunk Holes in Plates Subjected to Tension, Bending and Pin loading
Ref. 6 Shivakumar K., Bhagarva A.	2006	An Equation for Stress Concentration Factor in Countersunk Holes.
Ref. 7 TAE-T-NT-150104	A	Countersunk hole stress concentration
Ref. 8 Michael C.Y. Niu		Airframe Structural Design

STRESS CONCENTRATION FACTOR

Ref. 9	Thierry Stehlin	2003	Fatigue analysis of riveted or bolted connections using the finite element method
Ref. 10	ESDU 81006	1981	Stress concentrations factors axially loaded lugs with clearance-fit pins
Ref. 11	HSB 34112-02	B	Stress concentration factors for symmetrical isotropic lugs
Ref. 12	Warren C.Young	7 th	Roark's formulas for stress and strain
Ref. 13	Schijve J.	2009	Fatigue of structures and secondary bending in structural elements
Ref. 14	Chakherlou T.N.	2010	An investigation about interference fit effect on improving fatigue life of a holed single plate in joints

SOFTWARE

Program	Version	Description

Figure 30. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 23/03/2020	M.Carmen Blanquer Darío Fernández	First issue	All pages

Figure 31. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. Site	Getafe	Manching
Stress Analysis (TEASA)	Name DD/MM/AAAA	Name DD/MM/AAAA
Stress Methods (TEASP-TL3)	Name DD/MM/AAAA	Name DD/MM/AAAA

Figure 32. Approval signatures table for last revision of this chapter